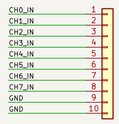
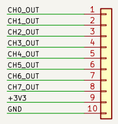
LOGIC ANALYZER / 8-CHANNEL INPUT PROTECTION

Topic 1 | RP2350 / Pico 2 | 3.3 V supply | Rev A – engineering prototype



J1

DUT INPUTS



J2

PICO INTERFACE



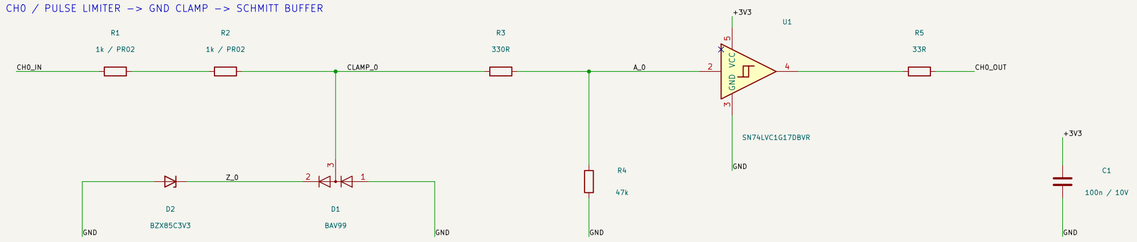




DESIGN CONDITIONS

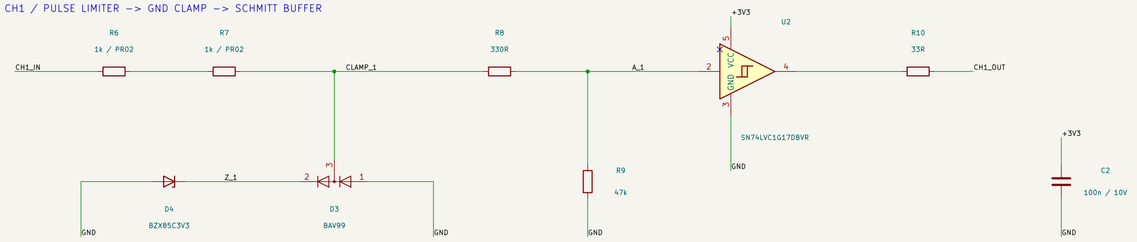
Target: +12 V DC, +200 V single rectangular pulse <= 1 ms.
Common DUT/Pico ground; NOT galvanically isolated.
Clamp current returns to GND, never to the +3V3 rail.
25 MS/s is the acquisition rate, not input bandwidth.
Validate 1 kHz...1 MHz first; characterize higher frequencies.
See README.md for calculations, limits and simulation status.

CH0 / PULSE LIMITER -> GND CLAMP -> SCHMITT BUFFER



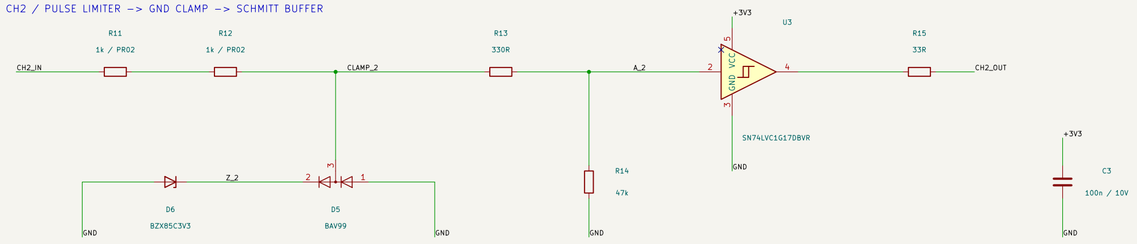
R1/R2 family: pulse-rated PR02, 2 W, 1%; 2k total per channel.

CH1 / PULSE LIMITER -> GND CLAMP -> SCHMITT BUFFER



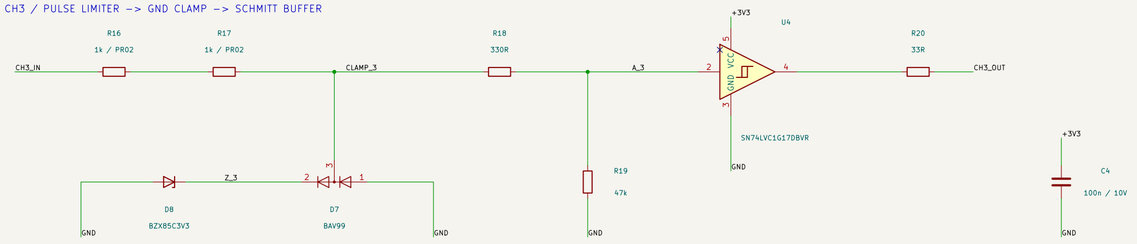
R1/R2 family: pulse-rated PR02, 2 W, 1%; 2k total per channel.

CH2 / PULSE LIMITER -> GND CLAMP -> SCHMITT BUFFER



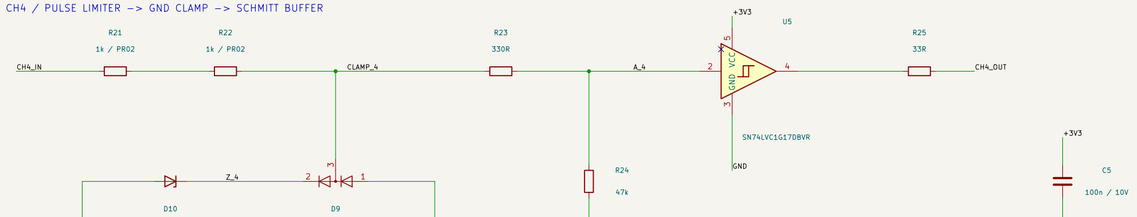
R1/R2 family: pulse-rated PR02, 2 W, 1%; 2k total per channel.

CH3 / PULSE LIMITER -> GND CLAMP -> SCHMITT BUFFER



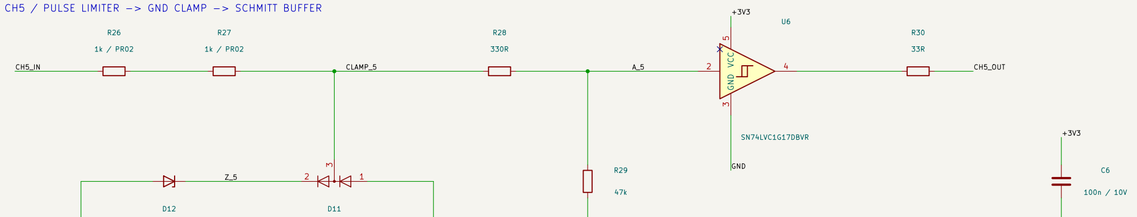
R1/R2 family: pulse-rated PR02, 2 W, 1%; 2k total per channel.

CH4 / PULSE LIMITER -> GND CLAMP -> SCHMITT BUFFER



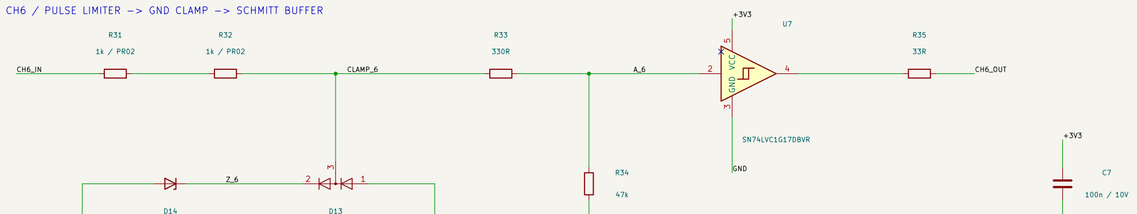
R1/R2 family: pulse-rated PR02, 2 W, 1%; 2k total per channel.

CH5 / PULSE LIMITER -> GND CLAMP -> SCHMITT BUFFER



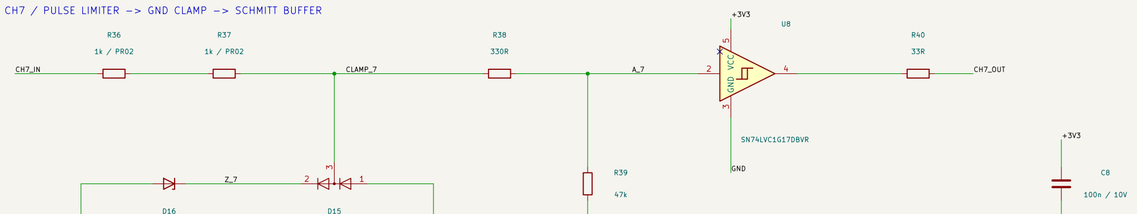
R1/R2 family: pulse-rated PR02, 2 W, 1%; 2k total per channel.

CH6 / PULSE LIMITER -> GND CLAMP -> SCHMITT BUFFER



R1/R2 family: pulse-rated PR02, 2 W, 1%; 2k total per channel.

CH7 / PULSE LIMITER -> GND CLAMP -> SCHMITT BUFFER



R1/R2 family: pulse-rated PR02, 2 W, 1%; 2k total per channel.

LAYOUT: keep BAV99/Zener return short; separate surge return from Pico ground path. Keep all eight input paths matched.

NO direct connection from CLAMP/A nodes to RP2350. U1...U8 outputs only. Do not fit SMBJ12CA across the signal path.

Prototype drawing: ERC checks connectivity only. Component models, parasitics and hardware tests determine the actual protection envelope.

Design target: +12V DC / +200V single pulse 1ms; see README for verification scope

Embedded Systems – Topic 1

Sheet: /

File: input_protection.kicad_sch

Title: 8-channel Logic Analyzer Input Protection

Size: A2	Date: 2026-11-05	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 1/1